Detecting Hidden Units and Network Size From Perceptible Dynamics

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Main question. A network has N interacting dynamical units, but only $n < N$ are measured. Can we infer the total size N—and hence the $h := N - n$ hidden units—without knowing the governing equations or network connections?

Main idea. Arrange observations from *several distinct transient trajectories* into a detection matrix T. Under sufficient excitation and observability, its rank equals the full system size: $\boxed{\operatorname{rank}(T) = N}$.

Sufficient Excitation: it means if we don not have enough different experiments or experiments have the same response then we can not detect the hidden units.

Observability: it means if the hidden units do not influence the measured units then we can not detect the hidden units.

1. Notation and dynamical model

N / n	Total number of units / number of directly measured units ($n < N$)
M / k	Number of distinct trajectories (experiments) / sampling times per trajectory
$\mathbf{z}(t) \in \mathbb{R}^N$	Full network state (including unmeasured variables)
$\mathbf{y}^{(m)}(t) \in \mathbb{R}^n$	Measurements from experiment $m = 1, \ldots, M$
$T \in \mathbb{R}^{kn \times M}$	Detection matrix assembled from measured trajectories

The unknown network evolves according to

$$\dot{\mathbf{z}}(t) = \mathbf{F}(\mathbf{z}(t)). \tag{1}$$

Near a fixed point $\mathbf{z}^*$, where $\mathbf{F}(\mathbf{z}^*) = 0$, let $\mathbf{x} = \mathbf{z} - \mathbf{z}^*$. A first-order approximation gives

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad A = D\mathbf{F}(\mathbf{z}^*), \quad \mathbf{x}(t) = e^{At}\mathbf{x}(0). \tag{2}$$

Here A is the (unknown) Jacobian, which describes local influences between units. Only n states are observed. With the selection matrix $C = [\, I_n \; 0 \,] \in \mathbb{R}^{n \times N}$,

$$\mathbf{y}^{(m)}(t) = Ce^{At}\mathbf{x}^{(m)}(0) = \Theta(t)\mathbf{x}^{(m)}(0), \qquad \Theta(t) = Ce^{At}. \tag{3}$$

The key point is that an observed response can depend on *all* N initial-state components, including those that are hidden.

2. Construct the detection matrix and derive the rank bound

At each common time t_j measured relative to the start of the experiment, arrange the M responses as columns:

$$Y(t_j) = [\mathbf{y}^{(1)}(t_j) \; \cdots \; \mathbf{y}^{(M)}(t_j)] \in \mathbb{R}^{n \times M}.$$

Stack the measurements at k times, preserving the same experiment in each column:

$$\boxed{T = \begin{bmatrix} Y(t_1) \\ \vdots \\ Y(t_k) \end{bmatrix}} = \underbrace{\begin{bmatrix} \Theta(t_1) \\ \vdots \\ \Theta(t_k) \end{bmatrix}}_{\mathcal{O}_k \in \mathbb{R}^{kn \times N}} \underbrace{[\mathbf{x}^{(1)}(0) \; \cdots \; \mathbf{x}^{(M)}(0)]}_{X_0 \in \mathbb{R}^{N \times M}}. \tag{4}$$

By the rank inequality for a matrix product,

$$\operatorname{rank}(T) \le \min\{\operatorname{rank}(\mathcal{O}_k), \operatorname{rank}(X_0)\} \le \min\{kn, N, M\} \le N. \tag{5}$$

If $\mathcal{O}_k$ has *full column rank* N (the chosen measurements reveal all state directions) and X_0 has *full row rank* N (the initial conditions span them), then

$$\boxed{\operatorname{rank}(T) = N, \qquad \widehat{h} = N - n = \operatorname{rank}(T) - n.} \tag{6}$$

For generic informative data, the paper uses $kn > N$ and $M > N$; these size inequalities alone do *not* guarantee full rank.

3. Practical algorithm: estimate rank using singular values

In measured data, noise makes exact numerical rank unreliable. The paper therefore computes the ordered singular values of T,

$$T = U\Sigma V^\top, \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_b \geq 0, \quad b = \min(kn, M),$$

and looks for an unusually large gap between successive values on a logarithmic scale:

$$\boxed{\Delta_j = \log \sigma_j - \log \sigma_{j+1} = \log\left(\frac{\sigma_j}{\sigma_{j+1}}\right), \quad j = 1, \ldots, b-1.} \tag{7}$$

The estimated network size is the *index* of the relevant gap, not its numerical height. The full iterative procedure is:

1. **Collect trials:** Start with $M = n + 1$ distinct transient trajectories, measured from the same n accessible units.
2. **Sample each trial:** Choose $k = \lceil M/n \rceil$ equally separated relative sampling times on each trajectory; construct T as in (4).
3. **Estimate a gap index:** Compute the singular values and Δ_j; among $j \geq n$, omit the potentially misleading indices $\{n, 2n, \ldots\}$ and $\{n+1, 2n+1, \ldots\}$. Identify the index of the largest remaining gap.
4. **Check robustness:** Repeat the sampling and gap calculation after retaining only $n-1, n-2, n-3, n-4$ measured units (when these counts are positive). Take the *median of the estimated indices* as $\widehat{N}^{(M)}$.
5. **Increase experiments:** Add another trajectory ($M \leftarrow M + 1$) and repeat until $\widehat{N}^{(M)}$ stops increasing. Report $\widehat{N} = \widehat{N}^{(M)}$ and $\widehat{h} = \widehat{N} - n$.

Intuition: Increasing M and k can reveal new independent directions. Once all N directions are represented, the rank estimate saturates; a spectral gap appears after σ_N. A noisy or temporarily flat estimate, however, is not by itself proof of saturation.

4. Assumptions and scope

- **Enough independent trials:** In the paper's generic full-rank argument, $M > N$ and $kn > N$. Distinct trials must excite enough independent initial-state directions; repeated identical trials are not sufficient.
- **Observability through measured units:** Hidden states must leave distinguishable signatures in the observed responses. A dynamically invisible unit cannot be counted by this procedure.
- **Comparable transient measurements:** Sampling times are aligned relative to each trial's start, and the underlying dynamics are consistent across trials.
- **Valid local linearization:** Near fixed points, (2) is a suitable approximation. For nonlinear periodic or chaotic motion, the method instead uses differences of *nearby trajectories*, $\delta\mathbf{z}(t) \simeq D\Phi_{t-t^*}|_{\mathbf{z}^*}\, \delta\mathbf{z}(t^*)$, so that a comparable linear relation holds locally. Trajectories must remain sufficiently close for this approximation.
- **Noise and conditioning:** Measurement noise, weak coupling, and poorly separated singular values can cause over- or underestimation. The exact-rank theorem does not guarantee a correct rank estimate on every finite, noisy dataset.

5. What did the authors find?

The paper demonstrates recovery of total network size in numerical experiments with (i) linear diffusively coupled systems near fixed points, (ii) nonlinear phase-locked Kuramoto oscillators, (iii) periodic and (iv) chaotic Rössler networks, and (v) nonlinear heterogeneous Michaelis–Menten biochemical networks with noise. In favorable examples, size detection worked when only about *10% of units were measured*. For tested linear networks, the minimum number of trials was consistent with $M_{\min} = N + 1$; in the shown Kuramoto experiments it was $N + 2$ because one trial served as a reference. More trials can allow accurate detection in noisy settings.

Final result and limitation. The method can infer the *number* of hidden units from sufficiently informative transient observations without first identifying the full governing equations or connectivity. It does *not* reconstruct hidden states, reveal where the hidden units are, or identify their exact connections.

Many Parts of this summary are generated by ChatGPT.

Source: Haehne *et al.*, *Phys. Rev. Lett.* **122**, 158301 (2019), especially Eqs. (1)–(13), Figs. 1–4 and the concluding discussion.